

Baishen Liang

Postdoctoral associate, Department of Neurology, Duke University
liangbs@mailbox.org; ORCID: [0000-0003-2604-9120](https://orcid.org/0000-0003-2604-9120)

EDUCATION

<i>Doctor of Philosophy in Cognitive Neuroscience</i> University of Chinese Academy of Sciences, Beijing, China	2017 – 2023 GPA: 3.94
<i>Exchanged Student, Department of Linguistics</i> University of California, Berkeley, Berkeley, CA, USA	2016 – 2016 GPA: 3.78
<i>Bachelor of Science in Psychology</i> Central China Normal University, Wuhan, China	2013–2017 GPA: 3.88

WORK EXPERIENCE

Postdoctoral Associate **July 2024 – present**
Department of Neurology, Duke University, Durham, NC, USA

- Lead a project to Design and establish an experiment to **study production-based verbal working memory maintenance and manipulation**. Collect, analyze, and maintain neurosurgical patients' intracranial electroencephalogram (iEEG) data.

Graduate Researcher **September 2017 – June 2023**
Institute of Psychology, Chinese Academy of Sciences, Beijing, China

Technology Consultant for the Core Facility IPCAS (Nominated Position)

- Assisted in maintaining transcranial magnetic stimulation (TMS) facilities.
- Gave regular public lectures and courses for faculty and students.
- Provided one-on-one technical consultation for faculty and students requiring TMS.

Project 1: Bilateral Laryngeal Motor Cortex in Speech Perception in Noise

- Lead a project to apply functional magnetic resonance imaging (fMRI) to localize Mandarin speakers' laryngeal motor cortices. Designed and established a TMS experiment, and collected and analyzed data.

Project 2: Transcranial Alternative Electric Stimulation (tACS) on Auditory and Motor Cortex

- Led a project to establish a technical pipeline combining phase-controlled dual-site tACS and electroencephalogram (EEG) recordings for auditory cognitive neuroscience research.

Project 3: Cortical Functional Map for the Perception of Speech Features

- Systematically reviewed 148 published fMRI studies on speech perception and comprehension.

PEER REVIEWED PUBLICATIONS

1. **Liang, B.**, Li, Y., Zhao, W., Du, Y.* (2023). Bilateral human laryngeal motor cortex in perceptual decision of lexical tone and voicing of consonant. **Nature Communications**, 14, 4710. [First author]
2. **Liang, B.** & Du, Y.* (2018). The functional neuroanatomy of lexical tone perception: an activation likelihood estimation meta-analysis. **Frontiers in Neuroscience**, 12, 495. [First author]
3. **Liang, B.** & Du, Y.* (2017). Mechanisms of sensorimotor integration in speech perception [言语知觉中的感觉运动整合机制]. **Science & Technology Review** [科技导报], 35(19):21-28. [First author]

MANUSCRIPTS UNDER REVIEW

1. Hu, W.#, **Liang, B.#**, Du, Y.* (2026). Singing enhances poetry learning and memorization in preschool children. **Annals of the New York Academy of Sciences (Revised and Resubmitted)**. [Co-first author]
2. Liu, J.#, Wang, Y.#, Feng, J.#, Zhao, H., He, Q., Yang, X.*, **Liang, B.* (Under review)** Successor representation supports structural learning of syllable sequences. [Co-corresponding author]
3. Wang, Y.#, Yu, K.#, Yin, S., **Liang, B.***, Wang, R.* (**Revision in progress**). Theta and gamma transcranial alternating current stimulation (tACS) modulate Mandarin consonant and lexical tone perception. (Preprint available on **bioRxiv**: 2025, 679546). [Co-corresponding author]

MANUSCRIPTS IN PREPARATION

1. **Liang, B.**, Earle-Richardson, A. M., Grant, G., Zafar, M., Frauscher, B., Southwell, D., Hickok, G., Cogan, G. B.* Sensory-motor mechanisms for verbal working memory. **In preparation**. [First author]

HONORS & AWARDS

- **Best Presentation Award in the 4th IEEE EMBS International Summer School of Neural Engineering.** Tsinghua University, Beijing, August 2018.
- **University of Chinese Academy of Sciences Student Grant (¥10,000).** University of Chinese Academy of Sciences, Beijing. December 2017.
- **National Scholarship (¥8,000).** Ministry of Education of the People's Republic of China, Beijing. November 2016.

PROFESSIONAL MEETINGS ATTENDED

- **Liang, B.**, Earle-Richardson, A. M., Grant, G., Zafar, M., Frauscher, B., Southwell, D., Hickok, G., Cogan, G. B.* Sensory-motor mechanisms for verbal working memory.
 - *Neuroscience 2025, Society for Neuroscience.* San Diego, CA, USA, November 2025 (poster).

- *Advances and Perspectives in Auditory Neuroscience*. San Diego, CA, USA, November 2025 (poster).
- *Society for the Neurobiology of Language 17th Annual Meeting*. Washington DC, USA, September 2025 (poster).
- *Duke Comprehensive Epilepsy Center Research Colloquium*. Durham, USA, September 2025 (slide presentation)
- *Duke Neurology TBS/DCEC Research Symposium*. Durham, USA, September 2025 (poster)
- *North Carolina Conference on Cognition 2025*. Chapel Hill, USA, April 2025 (poster).
- **Liang, B.**, Li, Y., Zhao, W., Du, Y.* Bilateral human laryngeal motor cortex in the perceptual decision of lexical tone and voicing of consonant (poster and presentation). *Society for the Neurobiology of Language Annual Meeting 2023*. Marseille, France, October 2023.
- **Liang, B.**, Hong, Y., Li, Y., Du, Y.* Causal involvement of auditory-motor low-frequency oscillation coupling in sentence perception under noisy conditions: a tACS study (poster). *Society for the Neurobiology of Language Annual Meeting 2023*. Marseille, France, October 2023.
- **Liang, B.**, Li, Y., Zhao, W., Du, Y.* A causal role of the human laryngeal motor cortex in lexical tone and voicing perception (poster and presentation). *Neurobiology of Language: Key Issues and Ways Forward II*. Nijmegen, Netherlands (online), March 2022.
- **Liang, B.**, Du, Y.* Neuroanatomy of lexical tone: an activation likelihood estimation meta-analysis (poster). *Organization for Human Brain Mapping Annual Meeting 2018*. Singapore, June 2018.

PEER REVIEW EXPERIENCE

Nature, 2025

Brain and Language, 2025

International Journal of Multilingualism, 2025

Acta Psychologica Sinica [心理学报], 2024

TECHNICAL SKILLS

- **Programming languages**
 - Matlab, Python, R, and C++.
- **Psychology**
 - Psychophysics experimental designs.
 - Statistics: linear models, logistic models, and drift diffusion models (DDMs).
- **Linguistics**
 - Acoustic processing and feature extractions by Praat/Matlab.
- **Neuroscience**
 - Electroencephalography (EEG), Magnetoencephalography (MEG), and intracranial EEG experimental design, data collection, preprocessing, and analyses.

- Transcranial magnetic stimulation (TMS) and transcranial electrical stimulation (tES) experimental design, data collection, and analyses.